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INTERNATIONAL ACADEMIC RESEARCH COMPETITION

Z-Cell Pore-Engineered Nanocarriers for Controlled Doxorubicin Delivery

Integrating Geometric Modeling, Spatiotemporal Computation, Physics-Informed Neural Networks, and Finite Element Analysis



PROPONENT:

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Introduction

Drug delivery describes the methods used to administer medications, balancing treatment efficacy with minimal side effects to maximize recovery [1]. Among delivery mechanisms, mesoporous silica nanoparticles (MSNs), due to their tunable structures, have drawn interest [2], enabling use in cancer treatment by encapsulating agents like doxorubicin [3], an effective anticancer drug [4]. However, MSNs face limitations including burst release, structural deformation, and poor targeting [5], harming healthy cells and failing to reach tumors even when the drug itself is effective [6]. Existing studies primarily focus on modifying surface chemistry [7], while optimizing nanoparticle geometry remains largely underexplored [8]. Recent mathematical discoveries have introduced soft cells, a new class of three-dimensional geometric structures called Z-cells seen in Figure 1, which exhibit unique topological properties usually found in nature [9], yet have not been investigated for nanocarrier designs.

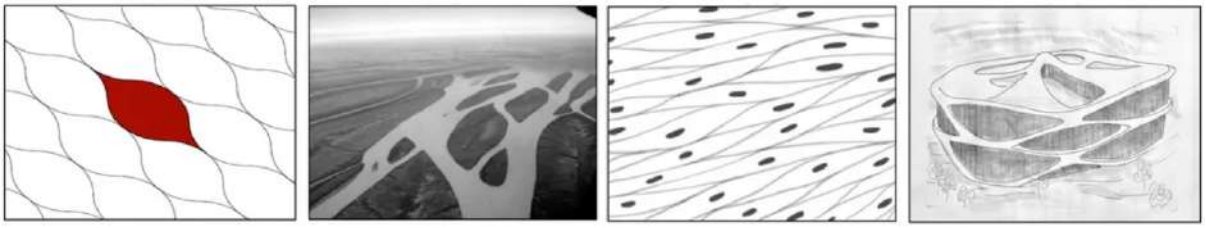


Fig 1. *Soft cell geometry in nature and design applications* [9].

Addressing these gaps, this study develops Z-cell pore-engineered nanocarriers for doxorubicin through geometric modeling and computational simulation. By introducing unexplored geometries into nanocarrier design, this study aims to demonstrate how shape can influence drug retention and release behavior. Furthermore, this study aims to achieve computational insights supporting the development of more efficient, stable, and controlled nanocarriers, advancing geometry-optimized drug delivery and demonstrating how mathematical discoveries can be translated into biomedical applications.

Research Questions

In line with the aforementioned, this study addresses the following research questions:

1. How do curvature, surface area, and packing efficiency of Z-cell structures influence their suitability as nanocarrier pore designs for drug delivery?
2. How does pore geometry affect drug retention, release behavior, and structural stability of nanocarriers?
3. How do Z-cell-based nanocarriers compare with conventional pore geometries based on predicted drug-delivery performance?

Literature Review

Prior work has examined MSNs for drug delivery, given that their porous structures can be tuned to accommodate therapeutic molecules and influence their release [10]. Their combination of high surface area, tunable pore volume, and modifiable surfaces allows drugs to be loaded into the nanopores and released in a controlled manner [11]. This is particularly

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relevant to doxorubicin (DOX), an anticancer drug whose use is often limited by toxicity and the need to control its distribution [12]. MSNs have been shown to improve DOX delivery through sustained or stimulus-responsive release, positioning them as an alternative to conventional administration [13]. Even so, how well these systems perform is affected by the carrier's structural design, and not just the material it is made of.

Pore size is one such characteristic known to affect drug loading and release [14]. For instance, one study found that varying MSN pore size altered the intracellular release of DOX, with larger pores producing faster release and greater intracellular accumulation [15]. From the same study comparing pore sizes of 2.3, 5.4, and 8.2 nm similarly found that pore size shaped drug loading, release behavior, cellular uptake, and anticancer activity, with the intermediate size performing best overall. Together, these findings suggest that larger pores do not automatically mean better delivery. Instead, pore dimensions must be optimized for the intended release profile, showing that geometry itself can be treated as a design variable rather than a fixed feature.

Despite these advances, most MSN literature centers on conventional pore structures, pore-size optimization, surface modification, or stimulus-responsive release [16]. Pore dimensions are well established as influencing DOX transport [17]. However, less is known about whether the shape of the pore, beyond its size, affects the drug's spatial and temporal behavior within a carrier. This creates an opportunity to examine alternative geometries, and one promising source of such geometries is biology itself, where structures have evolved to manage transport and flow within similarly confined, tube-like spaces [18].

The recently characterized "soft cell" is a class of space-filling shapes with minimal or no sharp corners, identified in biological structures such as muscle fibers and seashell chambers [9]. A specific subclass, the soft Z-cell, has been shown to tile prismatic tubes without gaps and has been associated with biological structures involving constrained, tube-like spaces. These Z-cell geometries tend to appear where biological systems must move material efficiently through narrow, channel-like confinements. This is a condition not unlike drug diffusion through a nanocarrier's pore network, suggesting that their curvature-dominant, low-corner form may offer similar transport advantages at the nanoscale. While existing MSN research has largely optimized pore size within conventional pore shapes, soft Z-cells provide an alternative geometry whose potential influence on drug transport remains insufficiently explored.

Evaluating such geometries requires computational methods capable of representing transport within complex structures. Finite element analysis (FEA) can model diffusion within defined geometries [19], while physics-informed neural networks (PINNs) take a different approach, incorporating equations as constraints during training so that model outputs remain consistent with established physical behavior [20]. Taken together, the literature supports an approach in which pore structure is treated as a key determinant of drug transport and computational modeling is used to compare alternative geometries. At the same time, existing research reveals limited investigation of biomimetic, low-corner pore geometries such as soft Z-cells for DOX delivery, particularly through combined FEA and

PINN approaches. This gap informs the present study's evaluation of Z-cell pore geometries, with emphasis on drug retention, controlled release, and the prediction of transport behavior over time.

Methodology:

Figure 2 illustrates the research processes to be conducted in this study, detailing the process from geometric analysis to model evaluation.

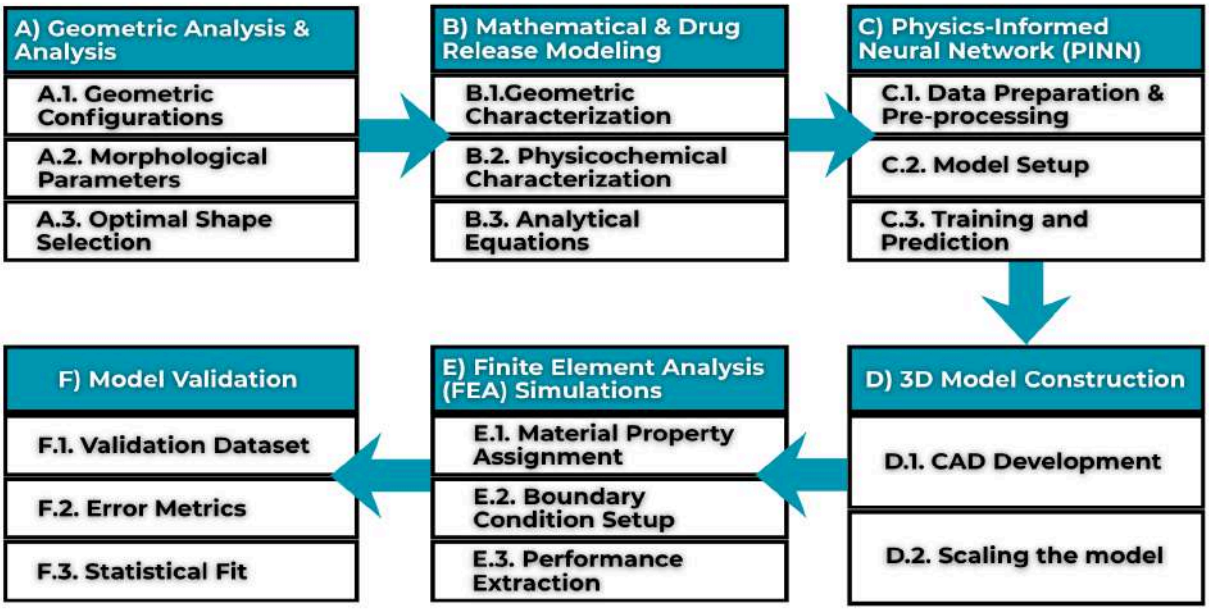


Fig 2. Flowchart of the Methodology

The study will use a quantitative approach to assess how different pore geometry affects mesoporous nanocarrier drug delivery.

1. **Geometric Characterization:** Four standard pore geometries, including spherical, cylindrical, hexagonal prism, and Z-cell, will be characterized using porosity, tortuosity, sphericity, aspect ratio, and surface-area-to-volume ratio.
2. **Drug Release Modeling and Optimization:** Doxorubicin loading and release dynamics will be modeled using effective diffusivity, diffusion length, and fractional drug release, followed by a weighted merit function to rank the pore geometries for controlled drug delivery suitability.
3. **Physics-Informed Neural Network:** A PINN built around the principles of Fick's Second Law will predict effective diffusivity and fractional drug release from the geometric characteristics described above.
4. **Model Validation:** The PINN's predictive accuracy will be checked against a held-out validation set, with RMSE and R^2 serving as the evaluation metrics.
5. **3D CAD Modeling:** Following computational optimization, the top-performing Z-cell geometry will be translated into a three-dimensional CAD model for structural analysis.

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6. **Finite Element Analysis:** The optimized geometry will undergo finite element analysis using amorphous silica material properties under simulated physiological and osmotic loading conditions to assess principal stress, von Mises stress, principal strain, and total deformation for structural stability.

Dataset:

The study will use a quantitative computational dataset consisting of 1,600 samples, with 400 samples generated for each pore geometry. The dataset will include geometric and drug-delivery parameters such as porosity, tortuosity, surface-area-to-volume ratio, effective diffusivity, and fractional drug release.

Sampling:

Stratified random sampling will be used to generate 400 computational samples for each pore geometry, with the dataset divided into 60% training and 40% validation sets.

Data Collection:

Data will be collected from published studies and scientific literature for pore geometry, doxorubicin properties, diffusion parameters, and material properties, while additional data will be generated through computational modeling, numerical simulation, PINN prediction, and FEA.

Data Analysis:

21 Prior to the main analysis, the dataset will be screened for a normal distribution using the Shapiro–Wilk test, and for equal variances across groups using Levene's test. Group differences across the four pore geometries will then be evaluated. Where parametric 23 assumptions are satisfied, this will involve a one-way analysis of variance (ANOVA) followed by Tukey's test for pairwise comparisons; otherwise, the Kruskal–Wallis test will be used as a non-parametric alternative. A weighted merit function will rank the geometries overall, PINN accuracy will be evaluated through RMSE and R^2, and mechanical stability will be assessed from the simulation's stress concentrations, strain values, and overall deformation.

Expected Outcome:

The study is expected to demonstrate the potential of Z-cell pore-engineered nanocarriers for improving DOX retention and controlled release compared with conventional pore geometries. It will evaluate the drug-delivery performance of different pore structures through geometric modeling, computational optimization, PINN prediction, and FEA. Results will identify the most promising Z-cell geometry and provide a computational basis for developing more controlled and structurally stable nanocarriers for DOX delivery.

Potential Limitation:

The study may be limited by the availability and accuracy of published parameters, and the static representation of drug diffusion and physiological conditions.

Conclusion:

Z-cell pore-engineered nanocarriers introduce a novel application of newly discovered mathematical structures to nanotechnology and drug delivery. By modeling Z-cell geometries as nanopores and evaluating their effects on drug transport, retention, and release, this study connects mathematical geometry with nanocarrier engineering. The resulting framework contributes new knowledge to mathematics by extending the application of the newly discovered Z-cell structures and to nanotechnology by providing a new geometry-based approach for designing nanocarriers for more controlled drug delivery.

Project Practicalities:

Figure 3 presents a Gantt chart outlining the schedule planned for this study.

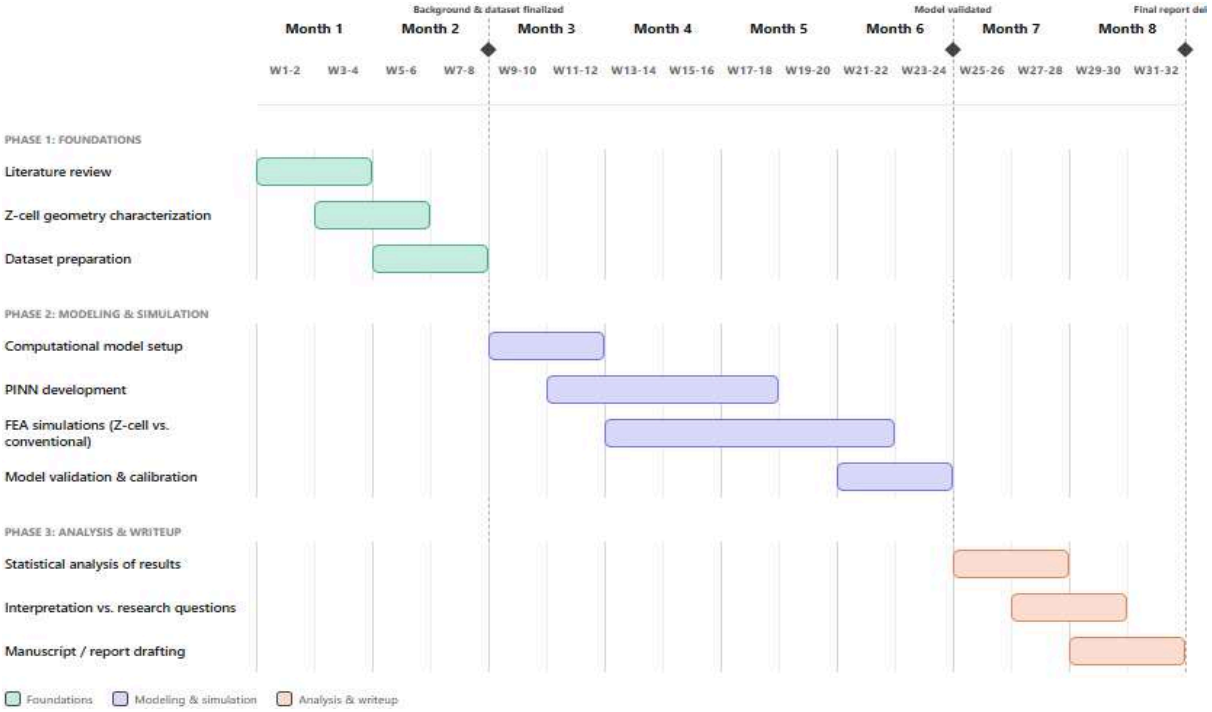


Fig 3. *Timeline of the proposed study.*

The project will span approximately 8 months. The first 2 months will cover literature review, Z-cell modeling, and dataset preparation. The next 4 months will focus on computational modeling, PINN development, and FEA. The final 2 months will cover statistical analysis, interpretation, and research presentation.

Post-Program Plan:

Future collaboration with relevant government agencies and research institutions, for instance: Department of Science and Technology (DOST) and molecular biology research groups at the University of the Philippines Diliman, may support experimental validation and further development of the nanocarrier. The findings may also be presented at academic research conferences such as Science Castle World or submitted to a relevant scientific journal, contributing to the continued application of Z-cell mathematical structures in nanotechnology and drug delivery.

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